INCH-POUND

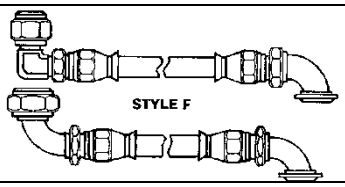
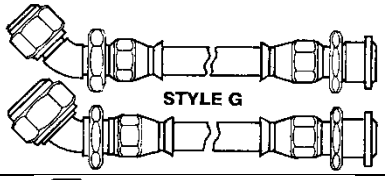
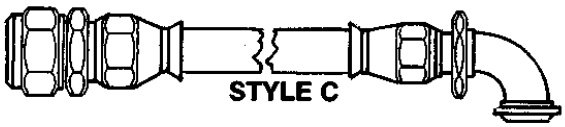
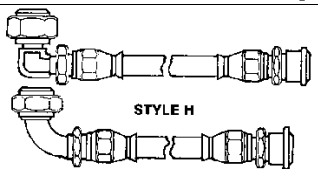
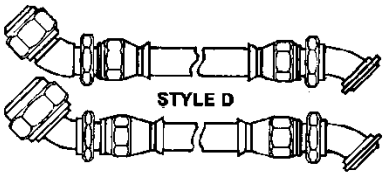
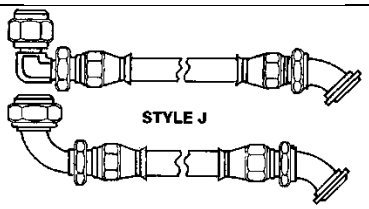
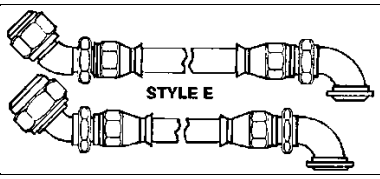
MIL-DTL-32434/4
23 January 2013

DETAIL SPECIFICATION SHEET

HOSE ASSEMBLY, POLYTETRAFLUOROETHYLENE,
CONVOLUTED, REUSABLE FITTINGS, HIGH TEMPERATURE,
MEDIUM PRESSURE, FLARELESS-TO-FLANGE

This specification is approved for use by all
Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-32434.

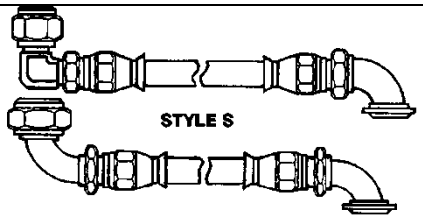
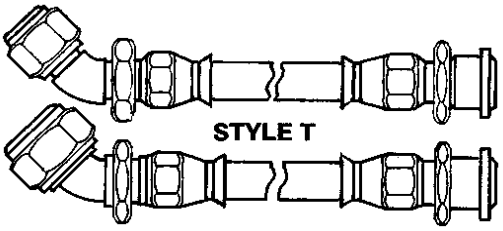
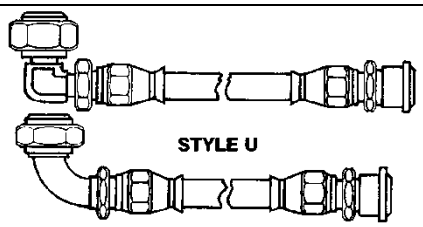
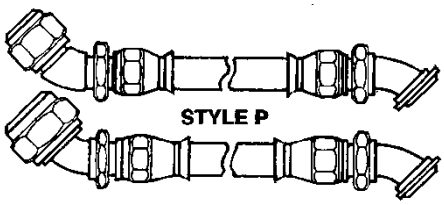
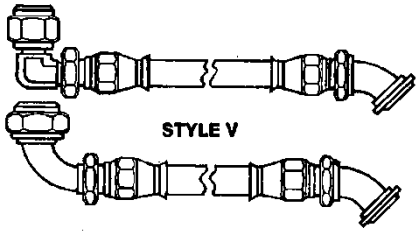
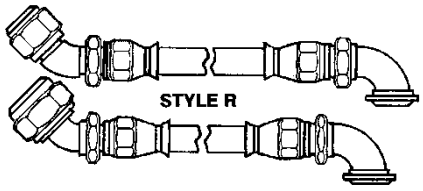
Hose assembly with class 1 fittings (450°F) corrosion resistant steel		Hose assembly with class 1 fittings (450°F) corrosion resistant steel	
End 1	End 2	End 1	End 2
			
			
			
			
			

NOTES:
1. See tables I and II for fitting ends and HCOF.
2. Fittings shall mate with parts designed to SAE-AS33514 and the mounting pad as shown on MS33786.

FIGURE 1. Class 1 hose assembly styles with flareless-to-flange reusable fittings.

AMSC N/A FSC 4720

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T06:21Z
Check the source to verify that this is the current version before use.

Hose assembly styles with class 2 fittings (275°F) combination of aluminum and corrosion resistant steel		Hose assembly styles with class 2 fittings (275°F) combination of aluminum and corrosion resistant steel	
End 1	End 2	End 1	End 2
			
			
			
			
			

NOTES:
1. See tables I and II for fitting ends and HCOF.
2. Fittings shall mate with parts designed to SAE-AS33514 and the mounting pad as shown on MS33786.

FIGURE 2. Class 2 hose assembly styles with flareless-to-flange reusable fittings.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T06:21Z
Check the source to verify that this is the current version before use.

Hose cut-off factor (HCOF) for assembly with class 1 fittings: See table I.

TABLE I. Hose cut-off factor (HCOF) for assembly with class 1 fittings. 1/

Hose assembly style with class 1 corrosion resistant steel (CRES) fittings 2/	Fitting ends		HCOF (size vs length) 3/					
	1	2	8	10	12	16	20	24
A	MS27381	MS27054	2.39	2.74	2.78	3.19	3.41	3.88
B	MS27381	MS27056	2.41	2.74	3.30	3.56	3.94	4.57
C	MS27381	MS27058	2.37	2.73	3.32	3.63	4.06	4.75
D	MS27384	MS27056	3.20					
	MS27382	MS27056		3.19	4.13	4.31	4.83	5.53
E	MS27384	MS27058	3.16					
	MS27382	MS27058		3.18	4.15	4.38	4.95	5.71
F	MS27385	MS27058	2.52					
	MS27383	MS27058		2.82	3.84	4.10	4.68	5.36
G	MS27384	MS27054	3.18					
	MS27382	MS27054		3.19	3.61	3.94	4.30	4.84
H	MS27385	MS27054	2.54					
	MS27383	MS27054		2.83	3.30	3.66	4.03	4.49
J	MS27385	MS27056	2.56					
	MS27383	MS27056		2.83	3.82	4.03	4.56	5.18

1/ Class 1 fittings, styles A through J, are made from CRES.
2/ For depiction of styles see figure 1.
3/ The HCOF is used in the following calculation to determine the hose length required to produce an assembly of a specific size, style and length: Assembly length - HCOF = Hose length. For example, the hose length required to produce an 30.00 inch length assembly of size 20, style F is calculated as follows: 30.00 - 4.68 = 25.32.

HCOF for assembly with class 2 fittings: See table II.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T06:21Z
Check the source to verify that this is the current version before use.

TABLE II. HCOF for assembly with class 2 fittings. 1/

Hose assembly style with class 2 aluminum and CRES fittings 2/	Fitting ends		HCOF (size vs length) 3/					
	1	2	8	10	12	16	20	24
K	MS27381	MS27054	2.39	2.74	2.78	3.19	3.41	3.88
M	MS27381	MS27056	2.41	2.74	3.30	3.56	3.94	4.57
N	MS27381	MS27058	2.37	2.73	3.32	3.63	4.06	4.75
P	MS27384	MS27056	3.20					
	MS27382	MS27056		3.19	4.13	4.31	4.83	5.53
R	MS27384	MS27058	3.16					
	MS27382	MS27058		3.18	4.15	4.38	4.95	5.71
S	MS27385	MS27058	2.52					
	MS27383	MS27058		2.82	3.84	4.10	4.68	5.36
T	MS27384	MS27054	3.18					
	MS27382	MS27054		3.19	3.61	3.94	4.30	4.84
U	MS27385	MS27054	2.54					
	MS27383	MS27054		2.83	3.30	3.66	4.03	4.49
V	MS27385	MS27056	2.56					
	MS27383	MS27056		2.83	3.82	4.03	4.56	5.18

1/ Class 2 fittings, styles K through V, are made from combination of aluminum and CRES.
2/ For depiction of styles see figure 1.
3/ The HCOF is used in the following calculation to determine the hose length required to produce an assembly of a specific size, style and length: Assembly length - HCOF = Hose length. For example, the hose length required to produce an 30.00 inch length assembly of size 20, style P is calculated as follows: 30.00 - 4.83 = 25.17.

REQUIREMENTS

Assembly classification: Class 1 and class 2 hose assemblies, as specified in MIL-DTL-32434, have been incorporated into the Part or Identifying Number (PIN) as a part of styles (see tables I and II and figures 1 and 2).

Dimensions: Unless otherwise specified, all dimensions are in inches.

Fittings: All fittings shall be qualified in accordance with MIL-DTL-27272.

Flareless fitting, hose connector design: NAS1760 flareless fitting end design see MIL-DTL-25579/1 for application and construction.

Hose: The hose shall be in accordance with MIL-DTL-32435.

Size code and elbow fitting drop height: See table III.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T06:21Z
Check the source to verify that this is the current version before use.

TABLE III. Size code and elbow fitting drop height.

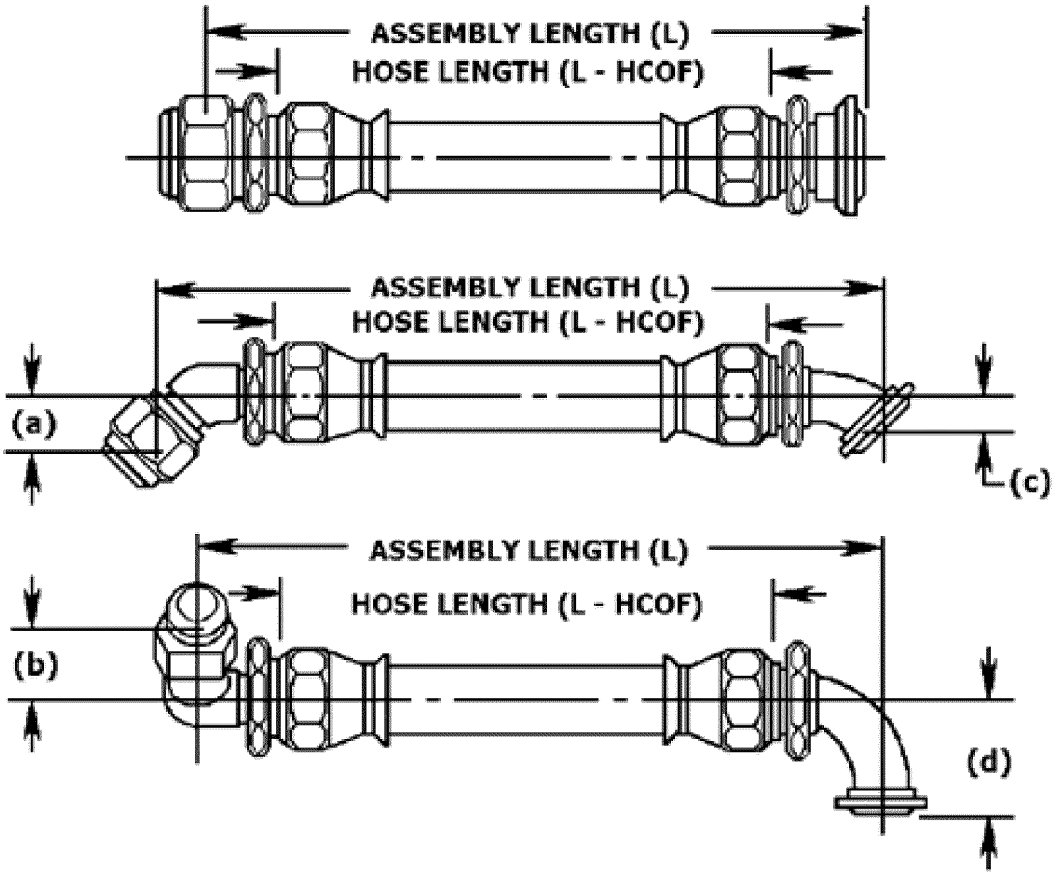
Size	Reference tube OD	Size code	Maximum drop height of elbow fitting			
			(a) 1/	(b) 1/	(c) 1/	(d) 1/
8	.500	H	.642	1.072	.458	.927
10	.625	J	.760	1.427	.410	.931
12	.750	K	.835	1.661	.503	1.191
16Z	1.000	M	.889	1.811	.540	1.317
20Z	1.250	N	.997	2.091	.605	1.535
24Z	1.500	P	1.190	2.473	.659	1.723

1/ Dimensions (a), (b), (c) and (d) are shown on figure 3.

Assembly length and tolerance: Hose assembly shall be furnished in lengths as specified in the contract or purchase order (see figure 4); however, tolerances on the length of each hose assembly shall be as follows:

a. ± 1/8 inch for lengths under 18 inches.
b. ± 1/4 inch for lengths from 18 inches to 36 inches.
c. ± 1/2 inch for lengths from 36 inches to 50 inches.
d. ± 1% for lengths over 50 inches.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T06:21Z
Check the source to verify that this is the current version before use.



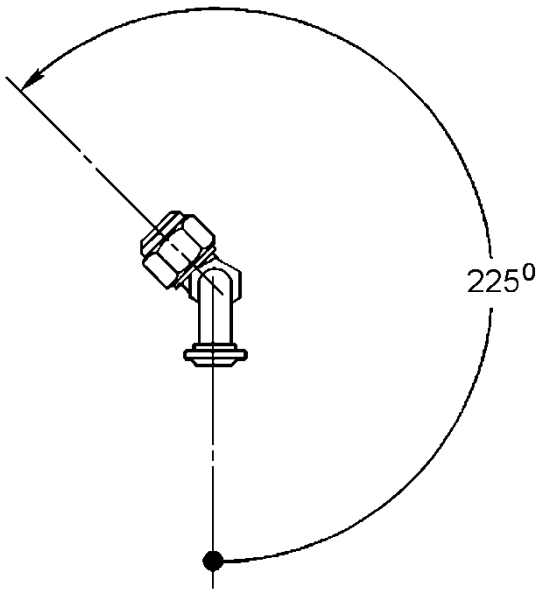
NOTES:

1. Hose assembly length “L” shall be measured, with the hose laid out horizontally and straight between the centers of the nipple end and the flange-sealing surface, along a straight line parallel to the hose length.
2. See table III for dimensions of (a), (b), (c) and (d). Dimensions (a) and (b) are measured to the end of a ball nose style fitting. Adjustments should be made for a NAS1760 style fitting see requirements application of NAS1760 and ball nose style fitting.

FIGURE 3. Elbow fitting drop height, dimensions (a), (b), (c) and (d).

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T06:21Z
Check the source to verify that this is the current version before use.

Angular alignment: Hose assemblies with elbow fittings on each end shall have the angular orientation between the elbows measured counter-clockwise from the centerline of the nearest fitting, positioned at six-o'clock, to the centerline of the other fitting (see figure 4). The elbow fitting drop height shall be as shown on figure 3 and table III. When applicable, the angular alignment shall be expressed in three digits and specified in the PIN.



NOTE: Angular alignment shall be measured in degrees with a tolerance of ±2°.

FIGURE 4. Measurement of angular alignment between elbow fittings.

Protective sleeve: If required, the hose assembly shall include a protective sleeve (see table IV) and its code shall be included in the PIN. Fire protective sleeve shall be subjected to testing in accordance with MIL-DTL-32434.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T06:21Z
Check the source to verify that this is the current version before use.

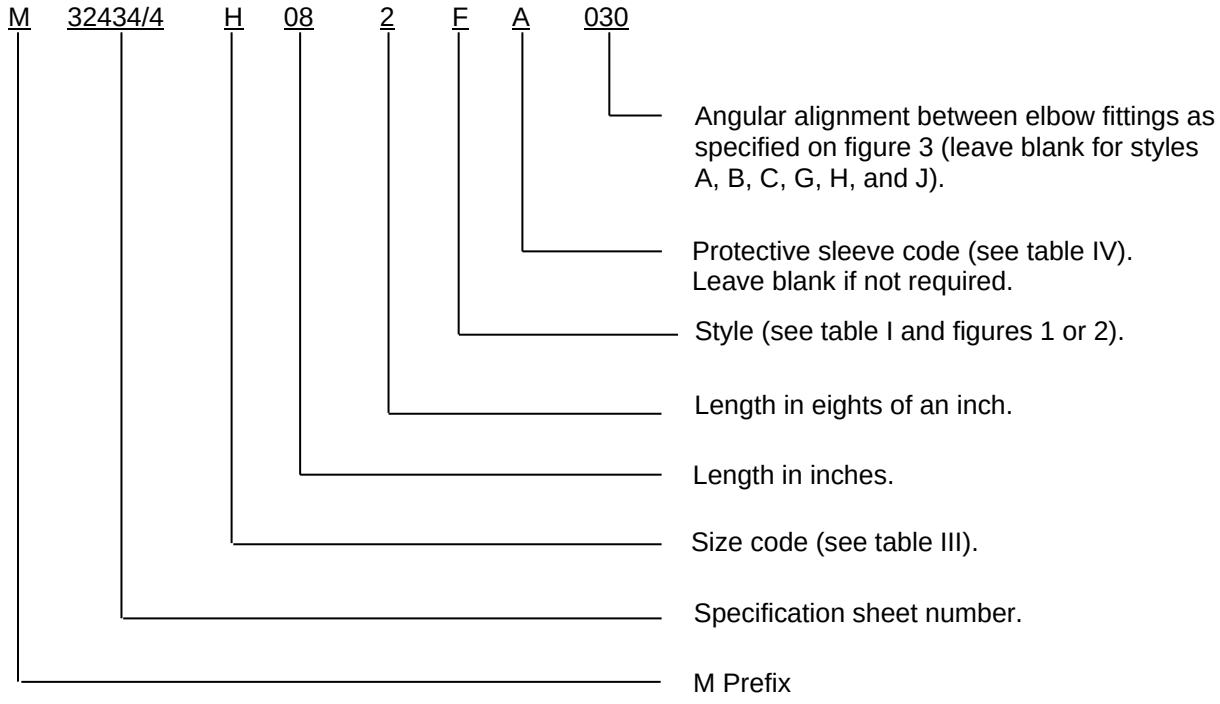
TABLE IV. Protective sleeve code.

Code	Type
Blank	None
A	SAE-AS1072 sleeve, fire protection, silicone covered, temperature ranging from -65°F to 500°F, secured with CRES bands as required. 1/
B	SAE-AS1073 - code B sleeve, abrasion protection, heat shrinkable, black polyolefin, temperature ranging from -65°F to 250 °F.
C	SAE-AS1291 - code A sleeve, chafe guard, extruded seamless white PTFE, temperature ranging from -65°F to 450°F, secured with CRES bands as required.
D	SAE-AS1291 - code C sleeve, chafe guard, extruded seamless transparent FEP, temperature ranging from -65°F to 350°F, secured with CRES bands as required.
E	SAE AS1298 sleeve, heavy wall chafe guard, extruded seamless black PTFE, temperature ranging from -65°F to 450°F, secured with CRES bands as required.
L	Lock-wire hole
F	Code A + L
G	Code B + L
H	Code C + L
J	Code D + L
K	Code E + L

1/ To prevent wicking of fluids, the cut end of the fire protective sleeve (code A) shall be coated with room temperature vulcanized (RTV) silicone rubber prior to installation. After installation, cracks and voids in the fire protective sleeve shall be coated with RTV rubber to prevent exposure of asbestos or fiberglass.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T06:21Z
Check the source to verify that this is the current version before use.

PIN: The PIN consists of letter M, specification sheet number, hose size code, hose length in inches and an eighths of an inch, hose assembly style, sleeve code, and angular alignment in degree between elbow fittings.



PIN example: M32434/4H082FA030 represents ½ inch OD PTFE convoluted hose assembly, 8.25 inches long, flareless-to-flange reusable fittings with fire protective sleeve and 30o angular alignment between elbow fittings.

Referenced documents. In addition to MIL-DTL-32434, this document references the following:

MIL-DTL-25579/1	MS27384
MIL-DTL-27272	MS27385
MIL-DTL-32435	MS33786
MS27054	NAS1760
MS27056	SAE-AS1072
MS27058	SAE-AS1073
MS27381	SAE-AS1291
MS27382	SAE-AS1298
MS27383	SAE-AS33514

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T06:21Z
Check the source to verify that this is the current version before use.

CONCLUDING MATERIAL

Custodians:
Army - AV
Navy - AS
Air Force - 99
DLA - CC

Preparing activity:
DLA - CC

(Project 4720-2012-022)

Review activities:
Army - AR, AT, EA, MI
Navy - MC, SA, SH
Air Force – 71

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T06:21Z
Check the source to verify that this is the current version before use.